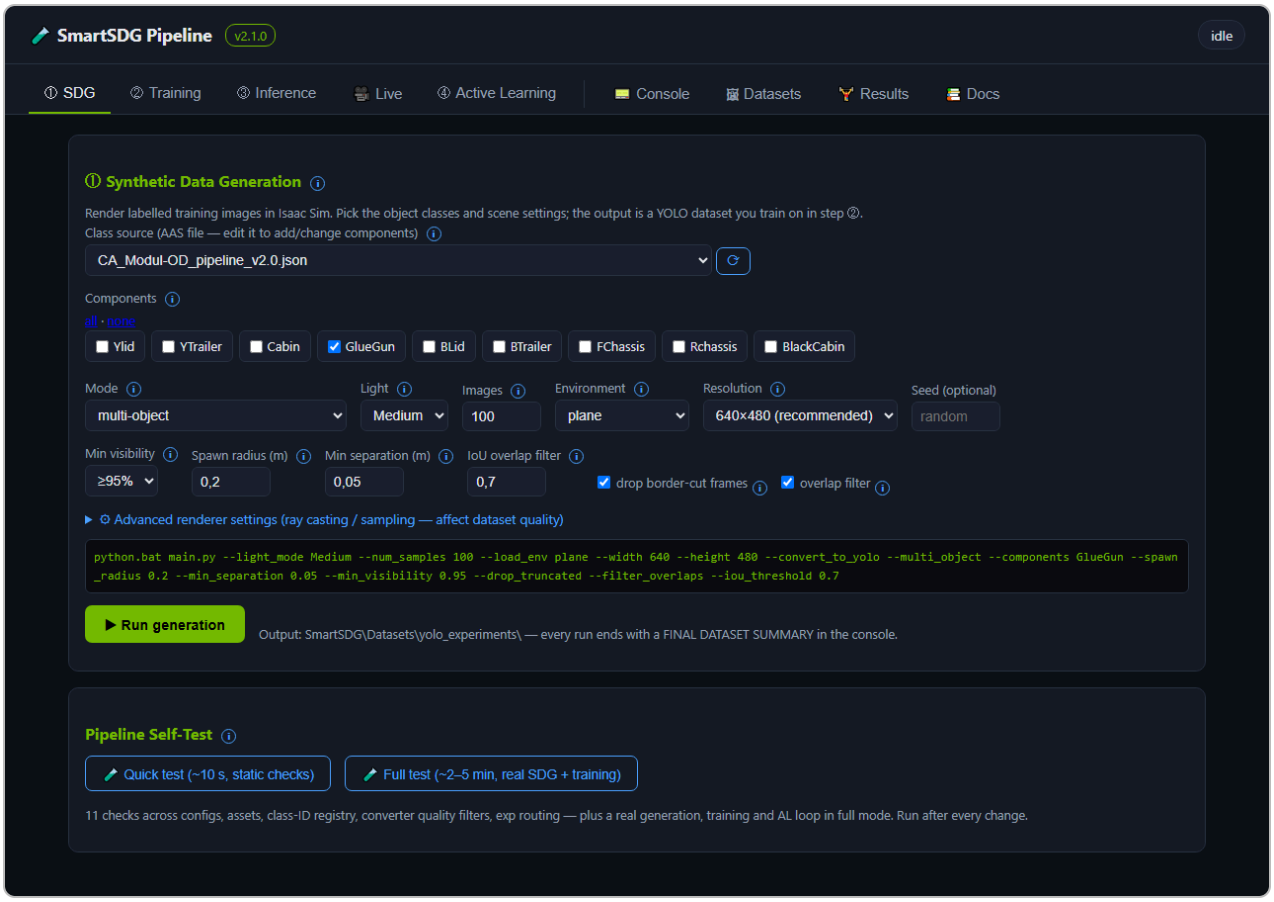
SmartSDG Pipeline

Comprehensive User Guide

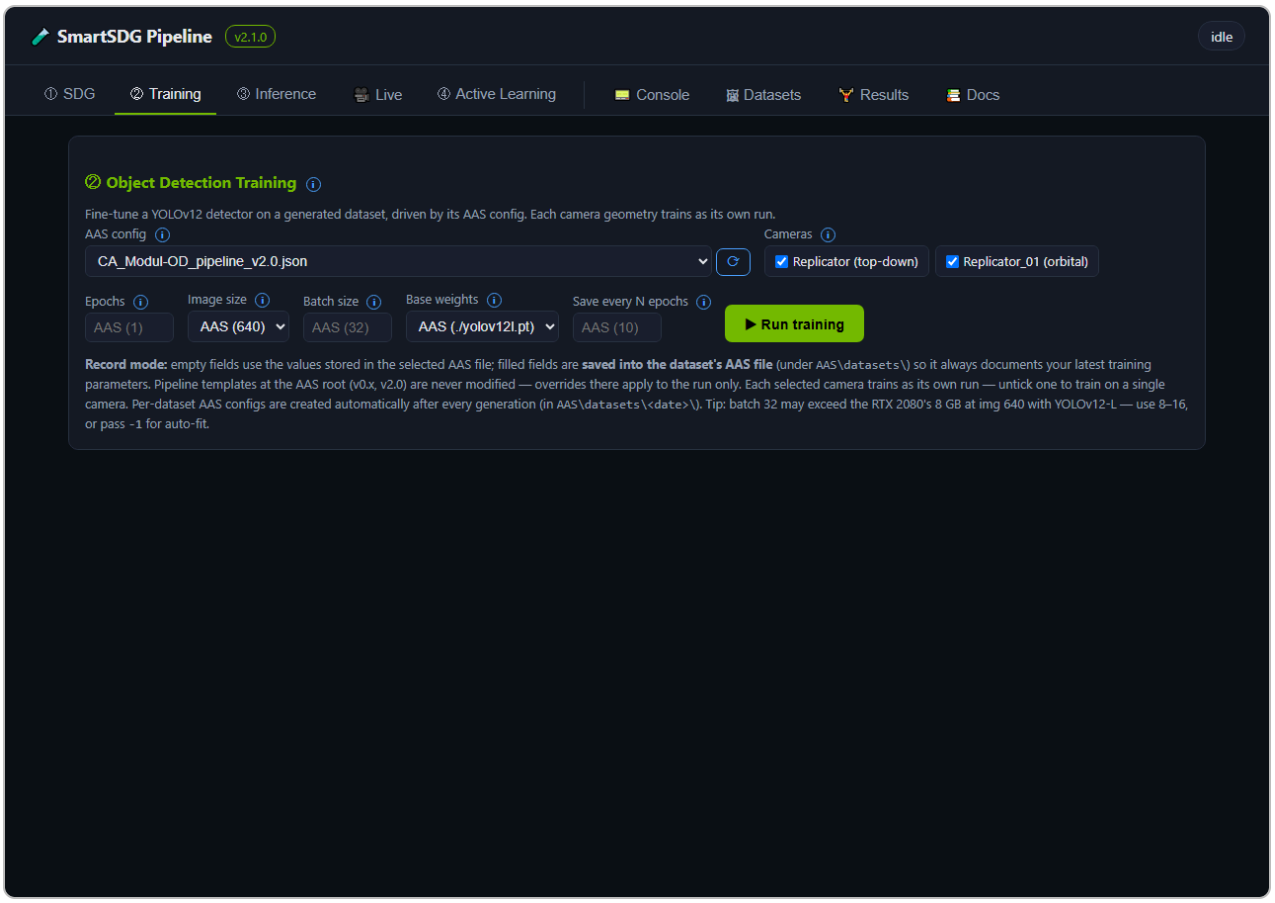
Synthetic Data Generation → Training → Inference → Active Learning

Version 2.1.0 · generated 2026-06-15

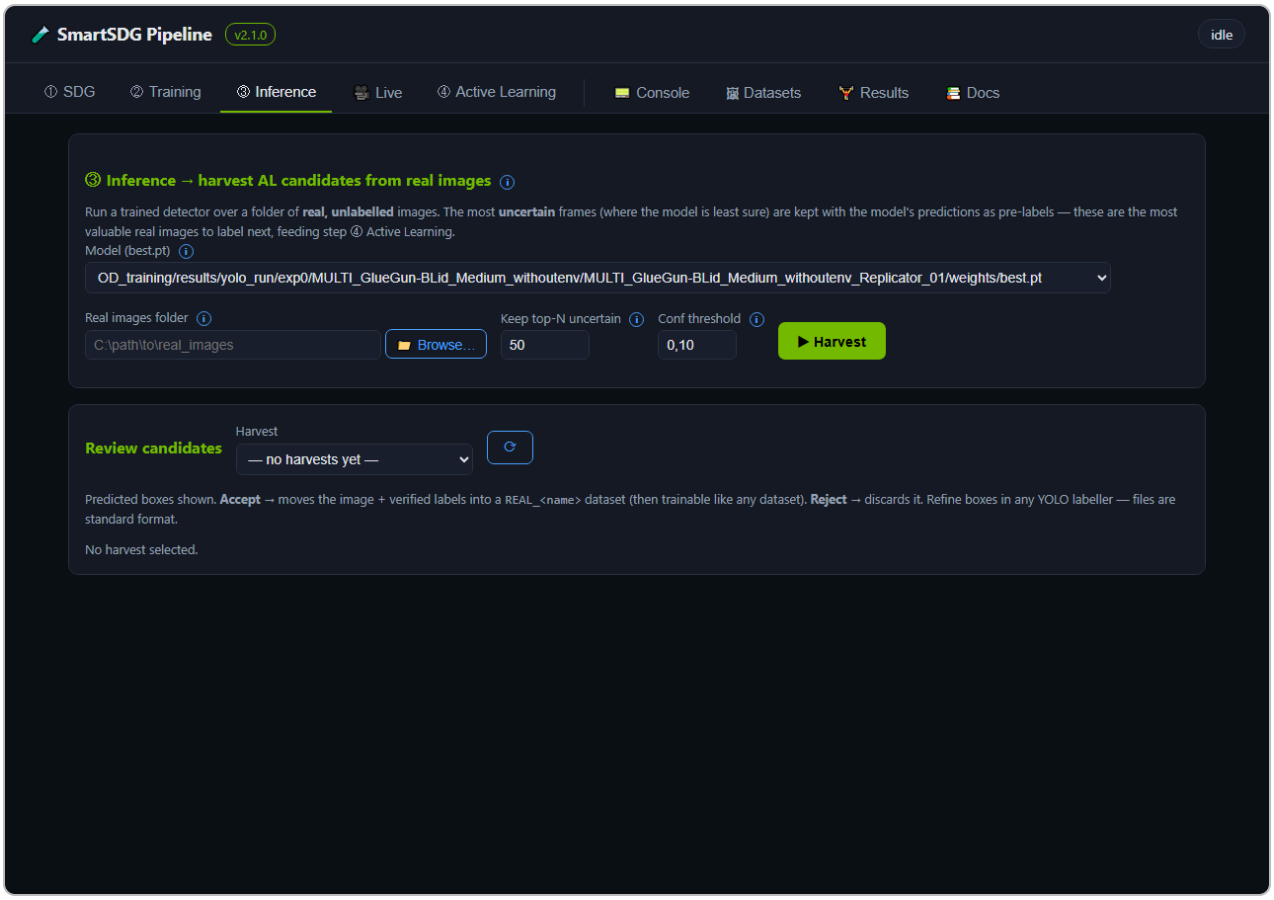
The dashboard at a glance



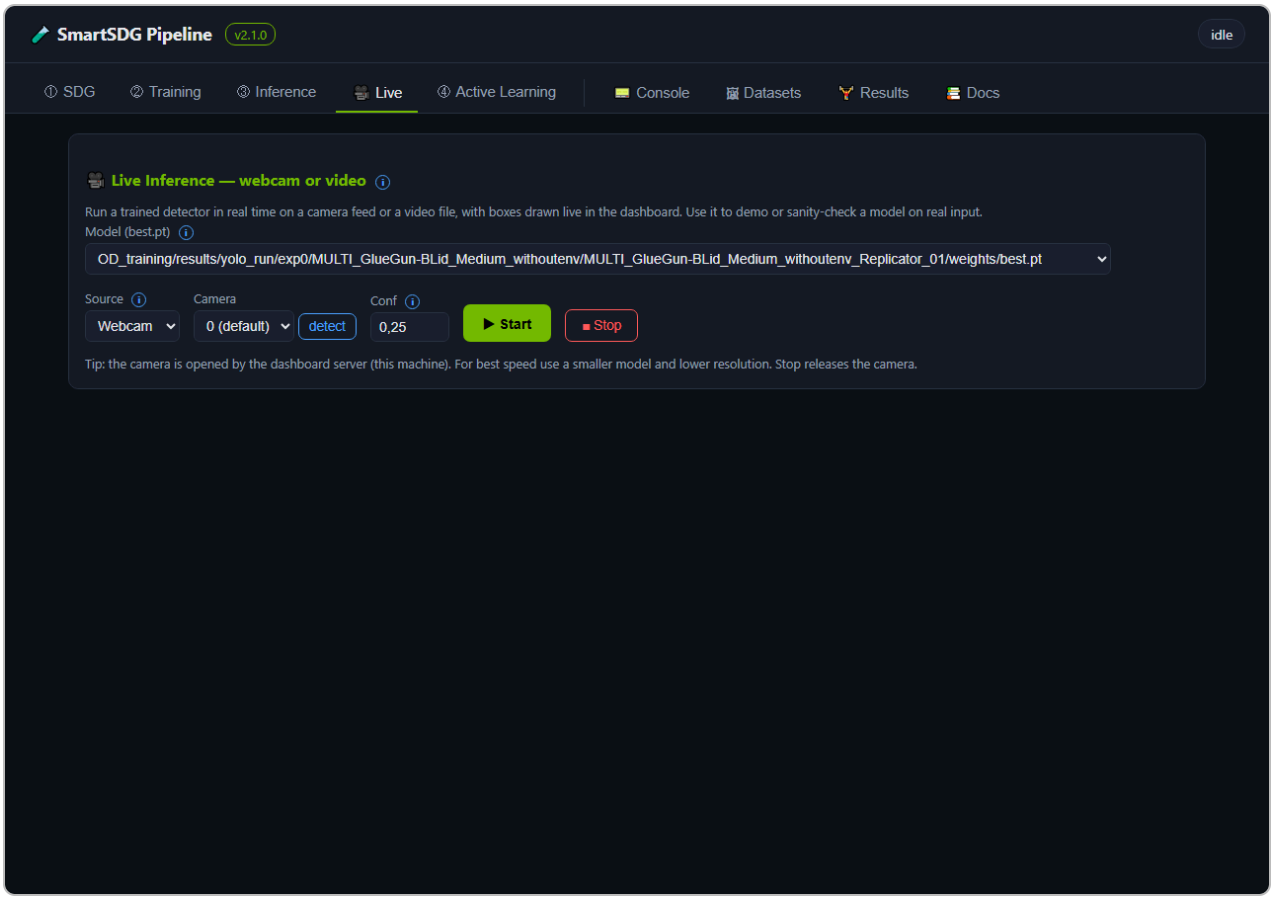
Synthetic Data Generation — pick components, scene and quality filters



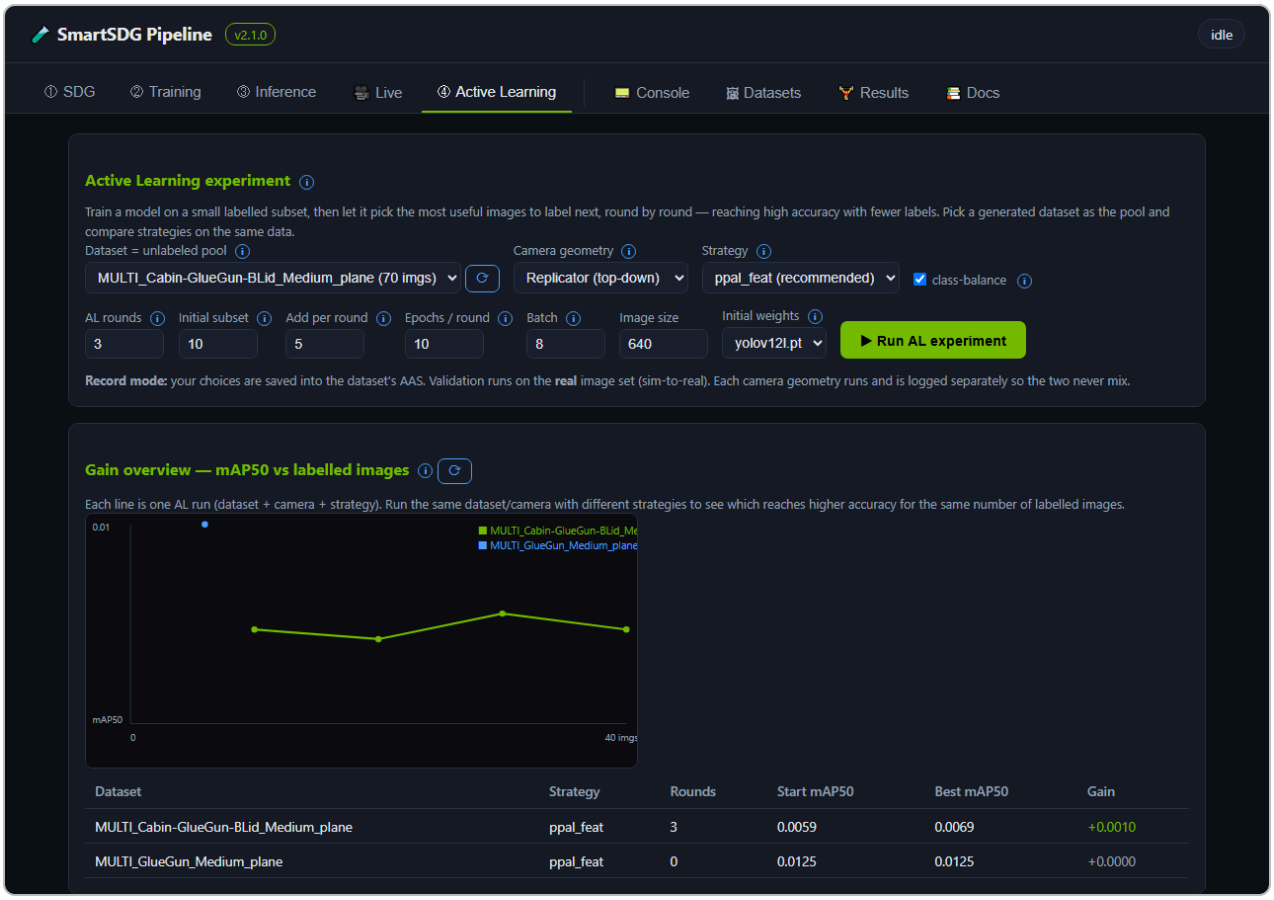
Object Detection Training — per-camera, record mode



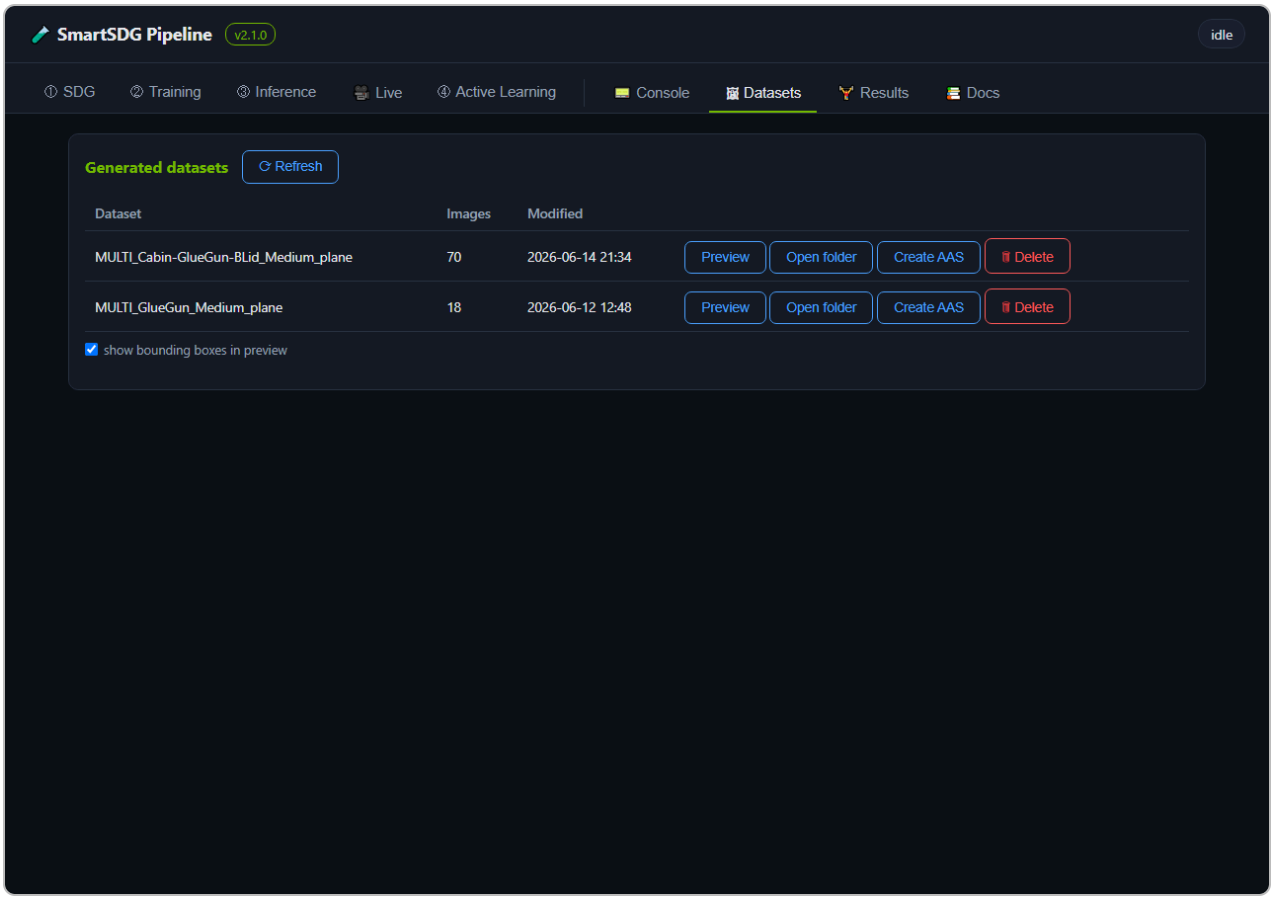
Inference — harvest the most-uncertain real images as AL candidates



Live Inference — webcam / video detection in real time



Active Learning — strategy, camera, budget, and the gain overview



Datasets — browse, preview boxes, edit labels, delete

SmartSDG Pipeline — Comprehensive User Guide

> A complete, example-driven manual for the whole pipeline, written for a professional user who is **not** expected to be an expert in synthetic data, object detection, or active learning. Every control in the dashboard has an ⓘ button whose "Open full guide" link brings you here.

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7. Step ④ — Active Learning
8. Labelling & the box editor
9. Where everything is stored
10. Reproducibility & the AAS record
11. Worked end-to-end example
12. Troubleshooting
13. Glossary

---

1. What this pipeline does

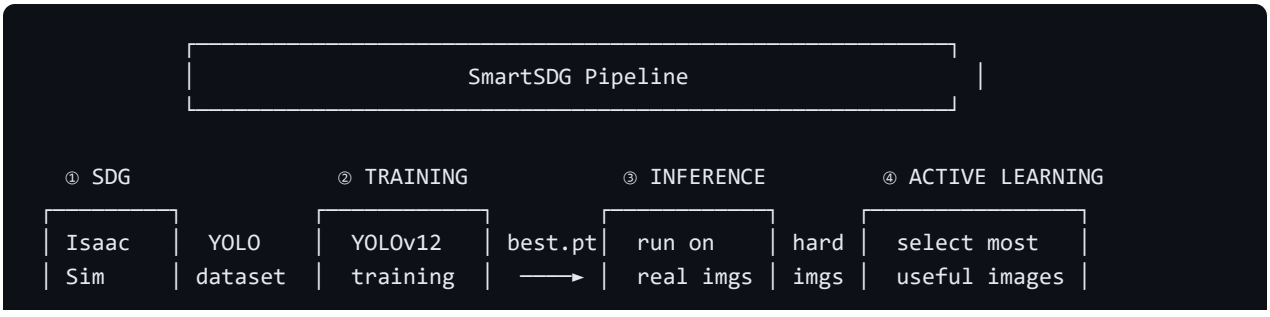
You want an object detector that works on **real** images of industrial truck components — but labelling thousands of real photos by hand is expensive. This pipeline lets you:

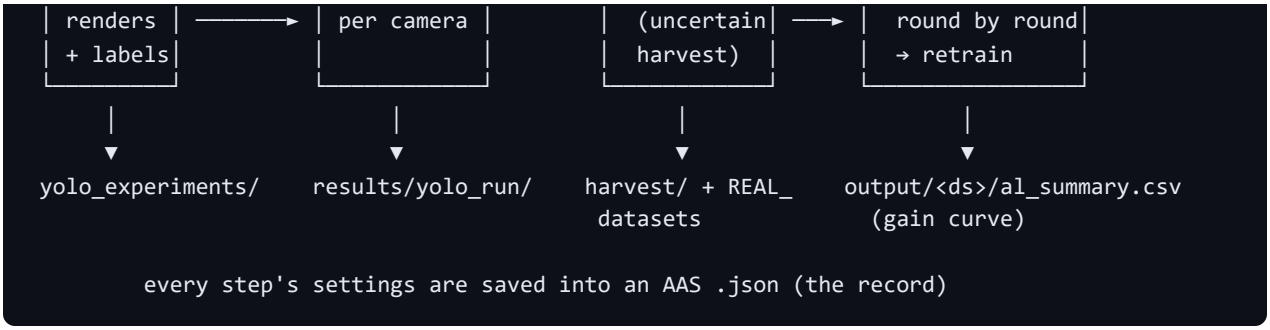
1. **Generate** perfectly-labelled training images in a 3D simulator (no photos, no manual labelling).
2. **Train** a YOLOv12 detector on them.
3. **Run** that detector on real images to find the ones it struggles with.
4. **Actively learn**: label only those most-useful images and retrain, reaching high accuracy with a fraction of the labelling effort.

Everything is driven from one local dashboard and recorded in AAS config files so any experiment is reproducible.

---

2. The big picture





Each scene is rendered by **two cameras** — `Replicator` (fixed top-down) and `Replicator_01` (random orbital) — and these are kept as **separate datasets** throughout, so you can train and compare each viewpoint independently.

---

3. Starting the dashboard

```
:: activate the training environment first, then:
scripts\launch_dashboard.bat
```

This opens `http://127.0.0.1:8765`. The top bar shows a **status badge** (idle / running / finished). The tabs follow the pipeline order:

```
① SDG | ② Training | ③ Inference | ④ Active Learning | Console | Datasets | Results | Docs
```

Any long-running job streams its output live to the **Console** tab.

---

4. Step ① — Synthetic Data Generation

Goal: produce a labelled YOLO dataset.

Controls

- **Class source** — the AAS file whose class list defines the objects and their global class IDs. To add a new object: add its name here and drop `<Name>.usd` into `SmartSDG/assets/Incorap_lkw/`; it auto-registers with the next free ID.
- **Components** — tick the objects to render. Grayed = missing USD file.
- **Mode** — `multi-object` (several parts per scene, for detection) or `single-object`.

- **Light / Images / Environment / Resolution / Seed** — scene settings. Seed makes a run fully reproducible (same object positions, rotations, lighting).
- **Quality filters** — min-visibility, spawn radius, min-separation, IoU overlap, drop border-cut. These guarantee every kept image contains only fully-visible, fully-labelled, non-overlapping objects (see HYPERPARAMETERS.md).
- **Advanced renderer** — samples-per-pixel, bounces, denoiser (render quality).

Example

Generate 100 multi-object images of three parts at 640×480 on a plane:

```
Components: Cabin, GlueGun, BLid
Mode: multi-object   Light: Medium   Images: 100   Environment: plane
Resolution: 640×480  Min visibility: ≥95%  Spawn 0.2  Separation 0.05
```

Press ▶ **Run generation**. The Console ends with:

```
##              FINAL DATASET SUMMARY                        ##
  Frames rendered by Isaac Sim       : 200
  Frames discarded - object occluded < 95% visible : 31
  FINAL images in dataset            : 142
  Output: ...\yolo_experiments\MULTI_Cabin-GlueGun-BLid_Medium_plane
```

The dataset now appears in the **Datasets** tab (with thumbnails + box preview).

---

5. Step ② — Object Detection Training

Goal: train a YOLOv12 detector on a generated dataset.

- **AAS config** — the dataset's config (auto-created after generation, listed at the top of the dropdown).
- **Cameras** — `Replicator` and/or `Replicator_01`. **Each ticked camera trains as its own separate run** — the two viewpoints never merge.
- **Epochs / Image size / Batch / Base weights / Save period** — standard training knobs. Empty fields use the AAS values; filled fields are saved back into the dataset's AAS (**record mode**).

> **Memory tip:** on an 8 GB RTX 2080, batch 32 at 640px with YOLOv12-L can run out of memory — use batch 8–16, or `-1` for auto-fit.

Press ▶ **Run training**. Results (weights `best.pt`, PR/F1 curves, confusion matrices) land in `OD_training/results/yolo_run/exp{0,1,2}/<dataset>/` and show up in the **Results** tab, one row per camera run, with a Weights ✓/— column.

---

6. Step ③ — Inference & harvesting

Goal: use a trained model to find the **real** images most worth labelling.

1. **Model** — pick a `best.pt` from your training runs.
2. **Real images folder** — click 📂 **Browse…** to navigate your drives and pick the folder (no need to type the path).
3. **Keep top-N uncertain** — how many of the hardest frames to keep.
4. **Conf threshold** — detection confidence used while scoring.

Press ▶ **Harvest**. The model scores every real image by uncertainty and keeps the top-N, saving each with its predicted boxes as **pre-labels**. They appear in the **Review candidates** grid with boxes drawn.

For each candidate: - ✏ **Edit** — open the box editor to correct the labels (see §8). - ✓ **Accept** — moves the image + labels into a `REAL_<name>` dataset (then trainable / AL-able like any dataset). - ✕ **Reject** — discards it.

This is the human-in-the-loop: the model tells you which real frames are worth your labelling time.

---

7. Step ④ — Active Learning

Goal: reach a target accuracy with far fewer labelled images, by letting the model choose what to label next. Full details in ACTIVE_LEARNING.md.

- **Dataset** = the unlabelled pool (a generated dataset).
- **Camera geometry** — run one camera at a time; each is logged separately so the two compare independently.

- **Strategy** — `random` (baseline) / `entropy` / `ppal_box` / `ppal_feat` (recommended).
- **Class-balance** — boosts rare classes.
- **Budget** — initial subset, rounds, add-per-round.

Each round trains a model, scores the pool, and adds the selected images. The **Gain overview** chart plots **mAP50 vs number of labelled images** for every run. To prove AL works, run the same dataset+camera with `random` and with `ppal_feat` and compare the two lines — the gap is the gain.

---

8. Labelling & the box editor

Click ✏️ **Edit** on any harvest candidate or dataset thumbnail to open the editor:

Action	How
New box	drag on an empty area
Select	click a box
Move	drag inside a selected box
Resize	drag the bottom-right corner handle
Delete	Del key, or the delete button
Change class	the class dropdown (applies to the selected/new box)

💾 **Save** writes standard YOLO labels (`class cx cy w h` , normalised). Files are plain text, so you can also refine them in LabelImg / Roboflow / CVAT.

---

9. Where everything is stored

What	Path
Synthetic dataset	`SmartSDG/Datasets/yolo_experiments/<name>/`
Real (accepted) dataset	`SmartSDG/Datasets/yolo_experiments/REAL_<name>/`
Raw render (pre-conversion)	`SmartSDG/Datasets/unified_render_output/<name>/`

What	Path
Training runs	`OD_training/results/yolo_run/exp{0,1,2}/<name>/`
Harvest candidates	`active_learning/harvest/<name>/{images,labels}`
AL outputs + gain curve	`active_learning/output/<name>[__<camera>]/al_summary.csv`
Config record (per dataset)	`AAS/datasets/<date>/CA_Modul-OD_dataset_<name>.json`

`exp0` = void background, `exp1` = plane, `exp2` = full 3D environment.

---

10. Reproducibility & the AAS record

Every dataset gets an **AAS** (Asset Administration Shell, IEC 63278) JSON file that records how it was generated and trained. The dashboard runs in **record mode**: when you change training or AL parameters, they are written into that file, so the dataset and the exact configuration that produced each result always travel together. These files are committed to git, giving a full history of every parameter change (`git diff`).

For exact run reproduction, set a **seed** in SDG — the scene geometry and labels become byte-identical across runs (rendered pixels match up to GPU floating-point noise, the standard caveat for any GPU renderer).

---

11. Worked end-to-end example

A complete cycle, all from the dashboard:

```
① SDG:        generate 100 images (Cabin, GlueGun, BLid), Medium, plane, 640×480
              → dataset MULTI_Cabin-GlueGun-BLid_Medium_plane (≈142 imgs after filtering)

② Training:   AAS = that dataset, camera = Replicator only, 50 epochs, batch 8,
              weights yolov12l.pt → best.pt in results/.../exp1/...

③ Inference:  model = that best.pt, Browse to your real-images folder,
              keep top 50, conf 0.10 → Harvest
              → review the 50 hardest real frames, Edit/Accept the good ones
              → creates dataset REAL_<folder>

④ AL:         dataset = the synthetic pool, camera = Replicator,
              strategy ppal_feat, 3 rounds, init 10, add 10
```

```
                → run again with strategy random
                → Gain overview shows ppal_feat above random = AL works
```

---

12. Troubleshooting

Symptom	Fix
`ModuleNotFoundError: aas_core3` (training)	activate the `yolov12` env before launching the dashboard, or `pip install aas-core3.0rc02`
Generation ends with `No valid data folders`	a config/parse error — run **Pipeline Self-Test** (Quick) to pinpoint it
`ParserError ... expected '<document start>'`	a config or AAS file has a UTF-8 BOM; the self-test "no BOM" check flags it
Training "Skipping … Path not found"	the AAS `base_data_dir` / camera names don't match the dataset folders
Out of GPU memory	lower batch size, or use `-1` for auto-fit; reduce image size
Dashboard shows "server offline"	the local server stopped — relaunch `scripts\launch_dashboard.bat`

Run `scripts\run_tests.bat` (or the **Pipeline Self-Test** buttons) after any change — it checks configs, assets, the class-ID registry, converter filters, exp routing, and (full mode) a real generation, training and AL loop.

---

13. Glossary

- **SDG** — Synthetic Data Generation (rendering labelled images in a simulator).
- **YOLO label** — a `.txt` per image; each line `class cx cy w h` (normalised).
- **Class ID** — the integer for each component, fixed by the registry order in `config.yaml`; never reordered so old datasets stay valid.
- **AAS** — Asset Administration Shell; the JSON config/record per dataset.
- **Camera geometry** — Replicator (top-down) vs Replicator_01 (orbital); kept separate.
- **Active Learning (AL)** — iteratively labelling the most informative images.

- **mAP50 / mAP50-95** — mean Average Precision, the standard detection accuracy metrics.
- **Uncertainty / diversity** — the two ingredients of AL selection: how unsure the model is, and how different the chosen images are from each other.
- **Record mode** — the dashboard saves your parameters into the dataset's AAS.

Active Learning — User Guide

Active Learning (AL) reaches a target accuracy with **far fewer labelled images** by letting the model choose what to label next, instead of labelling everything.

This guide explains every control in the dashboard's 🧠 **Active Learning** and 🎯 **AL Harvest** tabs. It is written for a professional user who is not an AL expert — every dashboard control also has an ⓘ button with a short hint that links here for the full explanation.

The idea in one paragraph

You train a model on a small labelled subset (round 0). The trained model then scores every remaining image in the pool, and a *selection strategy* picks the most useful ones to add to the labelled set. You retrain and repeat for a few rounds. Because the model labels its own "most informative" images first, the accuracy curve usually climbs faster than labelling at random — so you hit your target mAP with a fraction of the labels.

In this pipeline the **pool is your synthetic (SDG) dataset**, and **validation is on real images**, so the gain curve directly measures sim-to-real transfer.

Two ways to run AL

1. On synthetic data (🧠 Active Learning tab)

The selected generated dataset is treated as the unlabelled pool; AL picks the most informative synthetic images each round. Use this to compare strategies and camera

geometries on controlled data.

2. On real data via inference (🎯 AL Harvest tab)

Run a trained model over a folder of real, unlabelled images. The most *uncertain* frames are kept with the model's predictions as pre-labels. You review/correct them in the browser (✏️ Edit) and **Accept** them into a `REAL_<name>` dataset — which then becomes trainable / AL-able like any other. This is the human-in-the-loop loop: the model tells you which real frames are worth labelling.

Controls

Dataset = unlabelled pool

The generated dataset whose images form the pool to select from. Generate one in the **Run** tab first. Bigger pools give the strategy more to choose from.

Camera geometry

SDG renders every scene from **two cameras**: - **Replicator** — fixed top-down view - **Replicator_01** — random orbital view

These are different viewpoints and form **separate datasets**. A model trained on one geometry should be compared on that geometry, so pick **one camera** and run it; the run is logged separately (`output/<dataset>__<camera>`) and never mixed with the other. "both" merges them — only use it if you deliberately want a combined model.

Strategy

How the next images are chosen each round: | Strategy | What it does | When to use | |---|---|---| | **random** | uniform random pick | the baseline every method must beat | | **entropy** | most-uncertain images (model least confident) | uncertainty only | | **ppal_box** | uncertainty + diversity by bounding-box geometry | legacy behaviour | | **ppal_feat** | uncertainty + diversity by appearance embeddings | **recommended** |

`ppal_feat` uses the detector's own feature embeddings to avoid selecting many look-alike images — the most faithful to the PPAL paper and usually the best.

Class-balance boost

Real pools are often class-imbalanced. With this on, images containing under-represented classes get a selection bonus, filling in rare classes faster.

Budget: rounds / initial subset / add per round

- **Initial subset** — images labelled and trained on before the first selection (round 0 seed set). Too small and the first model is too weak to score well; a few dozen is typical.
- **AL rounds** — how many select+retrain cycles follow round 0.
- **Add per round** — images added each round (the labelling budget per step).

Constraint: `initial + rounds × add` must not exceed the pool size.

Training: epochs / batch / image size / initial weights

- **Epochs per round** — 3–10 for quick experiments, 50+ for real results.
- **Batch** — larger is faster but uses more GPU memory; on an 8 GB RTX 2080 use 8–16 at 640px with YOLOv12-L, or `-1` for auto-fit.
- **Initial weights** — the pretrained checkpoint round 0 fine-tunes from; **YOLOv12** is the project default.

Reading the Gain Overview

Each line is one AL run (dataset + camera + strategy), plotting **mAP50 vs number of labelled images**. The **gain** column is *best mAP50 − round-0 mAP50*.

To show AL works, run the **same dataset + camera** with `random` and with `ppal_feat`, then compare the two lines at the same number of labelled images — the vertical gap is exactly what Active Learning bought you. Compare the two camera geometries as separate lines to see which viewpoint transfers better.

Where everything is saved

What	Path
AL run outputs + learning curve	`active_learning/output/<dataset>[__<camera>]/al_summary.csv`
Harvest candidates (pending review)	`active_learning/harvest/<name>/{images,labels}`
Accepted real dataset	`SmartSDG/Datasets/yolo_experiments/REAL_<name>/`
AL parameters (record)	the dataset's AAS file under `AAS/datasets/<date>/`

Reproducibility: every AL run writes its parameters into the dataset's AAS file (record mode), so the config and the dataset always travel together.

SDG Hyperparameter Guide

Every knob that affects dataset quality, what it does, and the validated value. **Read this before changing anything** — SDG output is highly sensitive, and a single bad value silently degrades downstream OD training.

There are three groups, applied at different stages:

```
GROUP 1: scene placement  →  render  →  GROUP 2: quality filters  →  GROUP 3: renderer quality
(prevent problems)                       (discard bad frames)         (image fidelity/speed)
```

Group 1 — Generation-time (how the scene is BUILT)

These prevent problems before rendering. Nothing is deleted by them.

Parameter	Validated	Where	Effect
`--spawn_radius`	**0.2 m**	CLI / `config.yaml: multi_spawn_radius`	Max distance of objects from scene center. Smaller → objects stay inside camera view (no border cuts). Does

Parameter	Validated	Where	Effect
			not change object pixel size.
`--min_separation`	**0.05 m**	CLI	Minimum XY distance between object **centers** (collision-aware rejection sampling). Keeps objects from sitting in front of each other → prevents occlusion at the source. `0` disables.
`truck_scale`	0.01	`config.yaml`	World scale of components. **Never change** — 0.1 makes objects 10× too large/close.
`orbit_radius`	1.5	`config.yaml`	Camera 2 distance. Larger → smaller objects on screen (hurts OD). Prefer tuning `spawn_radius` instead.
`focal_length`	48.0	`config.yaml`	Camera lens. Larger → zoomed in (objects bigger, FOV narrower).
`multi_spawn_z_variation`	0.3	`config.yaml`	Vertical spread of spawned objects.
`LIGHT_INTENSITY_MAP`	Low/Medium/High	`config.yaml`	Mean/std lux per `--light_mode`. Keys MUST be `Low` / `Medium` / `High`.

Rule of thumb: if too many frames are being discarded by Group 2, fix it here (smaller spawn radius, larger separation), don't loosen the filters.

Group 2 — Filter-time (what is KEPT after rendering)

Applied during YOLO conversion. **v2 invariant: a filter never removes only a label — if any object fails, the whole frame is discarded** (an image with a visible-but-unlabeled object poisons OD training).

Parameter	Validated	Measures	Frame discarded when…
`--min_visibility`	**0.95**	Replicator `occlusionRatio` — real 3D occlusion **by other objects**	any object is > 5% hidden behind another
`--drop_truncated`	**on**	bbox touching the image border	any object is partially **outside the camera frame**
`--filter_overlaps` + `--iou_threshold`	**on, 0.7**	IoU of final YOLO **label rectangles**	two boxes overlap ≥ 0.7
(automatic)	always	degenerate zero-area boxes	an object exists but no valid box can be written

min_visibility vs filter_overlaps — not the same thing!

- `min_visibility` = true 3D occlusion (is the object hidden?). The scientifically meaningful one.
- `filter_overlaps` = 2D rectangle intersection. Two fully visible side-by-side objects can still have overlapping *boxes*. Coarse safety net only.
- With `min_visibility 0.95` active, the overlap filter rarely fires.

Policies

`--occlusion_policy` / `--truncation_policy`: `frame` (default — discard whole image) or `box` (remove only the label — **avoid**, leaves unlabeled objects).

Group 3 — Renderer / ray-casting quality (`config.yaml: SIMULATION_CONFIG`)

Parameter	Validated	Effect
`renderer`	PathTracing	Physically accurate. `RayTracedLighting` = faster, less realistic.
`samples_per_pixel_per_frame`	8	Rays per pixel. Main quality/speed knob. 32–64 for final datasets, 8 for iteration (denoiser compensates).
`max_bounces`	32	Light path depth.
`max_specular_transmission_bounces`	6	Bounces through glass/specular.
`max_volume_bounces`	4	Volumetric media.
`denoiser`	True	OptiX AI denoiser — lets low spp look clean.

Advanced carb settings (firefly filter, totalSpp, DLSS mode, RTSubframes…) go in `config.yaml: RTX_CARB_SETTINGS` — applied automatically before the Replicator graph builds. Commented examples are in the config file.

Validated baseline command (2026-06-10)

```
C:\Users\hota02\Documents\omniverse\python.bat main.py ^
  --config "src\config\config.yaml" ^
  --multi_object --components Ylid,YTrailer,Cabin,GlueGun,BLid,BTrailer,FChassis ^
  --light_mode Medium --num_samples 100 --load_env plane ^
  --width 640 --height 480 ^
  --convert_to_yolo --filter_overlaps --drop_truncated
```

(min_visibility 0.95, policies frame, IoU 0.7, spawn 0.2, separation 0.05 are defaults — only the switches need passing.)

How to change a hyperparameter safely

1. Change **one** parameter.
2. Generate a small batch (`--num_samples 20`).
3. Inspect: images visually + the FINAL DATASET SUMMARY (discard rates).
4. Record old → new value and observed effect in `CHANGELOG.md` .

5. Only then scale up to full generation.

Running on Another Machine — Portability & Docker

TL;DR

The pipeline no longer hard-codes any machine path. On a new machine you either:

1. **Set three environment variables**, or
2. **Copy** `pipeline_config.example.json` → `pipeline_config.json` and fill in the paths, or
3. **Do nothing** — auto-detection finds standard install locations.

Check what got resolved at any time:

```
python config_paths.py
```

```
  isaac_python   = ...\omniverse\python.bat            [OK]
  train_python   = ...\envs\yolov12\python.exe         [OK]
  val_real       = ...\val_real                        [OK | MISSING]
```

The three paths:

Key	What	Env var
`isaac_python`	Isaac Sim's `python.bat` (SDG only)	`SMARTSDG_ISAAC_PYTHON`
`train_python`	Python with torch+ultralytics+aas_core3 (OD/AL/CL/inference)	`SMARTSDG_TRAIN_PYTHON`
`val_real`	folder of real validation images (optional)	`SMARTSDG_VAL_REAL`

Resolution order per path: **env var** → `pipeline_config.json` → **auto-detect** → **fallback**. `pipeline_config.json` is git-ignored so each machine keeps its own.

Can Isaac Sim 4.5 run in Docker? — honest answer

Partly, and not on this Windows setup. Here is the real situation so you can decide:

- NVIDIA *does* publish an Isaac Sim **Docker image**, but it is **Linux-only**, ~20 GB, and requires the **NVIDIA Container Toolkit** with GPU passthrough.
- Your current SDG runs on **Windows-native Isaac Sim** (the way NVIDIA ships it for Windows). Windows Isaac Sim does **not** containerize; the Docker image would only run under **WSL2 + GPU passthrough** (Linux inside Windows), which is a large, fragile setup and would not reuse your existing working install.
- So: **Dockerizing the SDG stage on Windows is not worth it.** The native install already works, and the path-config change above already makes it portable to any Windows machine with Isaac Sim installed.

What *is* worth containerizing: the Python stages. OD training, Active Learning, Continual Learning and inference are plain PyTorch/Ultralytics — they containerize cleanly and run on any machine (Linux or Windows+WSL2) with an NVIDIA GPU. A `Dockerfile` for that is provided (see below). The split is:

```
SDG (Isaac Sim)            →  native install (Windows or Linux), path-configured
OD / AL / CL / Inference   →  Docker (portable) OR the same native conda env
```

Docker for the Python stages (optional)

`docker/Dockerfile` builds an image with torch (CUDA), ultralytics and aas_core3. Datasets, weights and configs are mounted as volumes, so the container is stateless.

```
# build
docker build -t smartsdg-train -f docker/Dockerfile .

# train (Linux host with nvidia-container-toolkit)
docker run --gpus all -v "$PWD:/workspace" -w /workspace/OD_training \
    smartsdg-train python main.py --mode train \
    --aas_file ../AAS/datasets/<date>/<config>.json
```

This does **not** include Isaac Sim — generate datasets natively first, then train/AL/infer in the container (or just keep using the conda env; the container is only for moving to a clean GPU box).

Recommended setup on a fresh machine

1. Install **Isaac Sim 4.5** (native) — only needed for SDG.
2. Create the Python env: `pip install -r requirements.txt` (or build the Docker image).
3. `python config_paths.py` — confirm all three paths resolve `[OK]`; if any are `MISSING`, set the env var or edit `pipeline_config.json`.
4. `scripts\run_tests.bat` — the self-test confirms the whole chain.
5. `scripts\launch_dashboard.bat`.